

CS502 MIDTERM PREPARATION

LEC WISE MCQS from lec 1 to 10

Solved by Anie

1. **The term "Algorithm" is derived from:**
 - A) Euclid
 - B) Turing
 - C) Al-Khwarizmi ☐
 - D) Fibonacci
2. **An algorithm is:**
 - A) A software
 - B) A hardware component
 - C) A set of programming instructions
 - D) A well-defined computational procedure ☐
3. **What distinguishes an algorithm from a program?**
 - A) Syntax
 - B) Efficiency
 - C) Independence from language/hardware ☐
 - D) Memory usage
4. **Macro programming issues relate to:**
 - A) Performance
 - B) Coordination of multiple programmers ☐
 - C) Loop optimization
 - D) Variable names
5. **Micro issues are focused on:**
 - A) Team management
 - B) High-level architecture
 - C) Critical code sections that consume most execution time ☐
 - D) Hardware design
6. **Worst-case time complexity is:**
 - A) Best possible performance
 - B) Maximum time taken for input size n ☐
 - C) Minimum time required
 - D) Random average
7. **Which of the following is NOT a resource considered in algorithm analysis?**
 - A) Time
 - B) Space
 - C) Battery life ☐
 - D) Bandwidth

8. **Average-case analysis depends on:**

- A) Sorting
- B) Probability distribution of inputs ☐
- C) Execution environment
- D) Code length

9. **The primary mathematical model used is:**

- A) CPU architecture
- B) RAM model ☐
- C) Neural network
- D) Turing machine

10. **Which algorithm was used to solve 2D Maxima problem in brute force?**

- A) Merge sort
- B) Plane sweep
- C) Nested loops
- D) Double for loop comparing all points ☐

11. **Worst-case time for brute-force maxima algorithm is:**

- A) $\Theta(n \log n)$
- B) $\Theta(n^2)$ ☐
- C) $\Theta(n)$
- D) $\Theta(\log n)$

12. **What does Θ notation represent?**

- A) Upper bound
- B) Lower bound
- C) Exact growth rate ☐
- D) None of the above

13. **What is the sum of first n natural numbers?**

- A) n
- B) $n(n+1)/2$ ☐
- C) n^2
- D) $\log n$

14. **Which of the following is a geometric series?**

- A) $1 + 2 + 3 + \dots$
- B) $1 + x + x^2 + \dots + x^n$ ☐
- C) $1 + 1/2 + 1/3 + \dots$
- D) $1 + 2^2 + 3^2$

15. **The harmonic series grows as:**

- A) Constant
- B) Linear
- C) Logarithmic ☐
- D) Quadratic

16. **Running time of NESTED-LOOPS example was:**

- A) $\Theta(n^2)$

- B) $\Theta(n \log n)$
C) $\Theta(n^2)$ ☐
D) $\Theta(n^4)$
17. **Summation of i^2 from $i=1$ to n is:**
A) $n(n+1)/2$
B) $(n(n+1)(2n+1))/6$ ☐
C) $2n$
D) $n!$
18. **Which function grows fastest asymptotically?**
A) $\log n$
B) n
C) $n \log n$
D) 2^n ☐
19. **In asymptotic notation, $O(g(n))$ represents:**
A) Lower bound
B) Exact value
C) Upper bound ☐
D) Average
20. **Which of the following is a valid example of $\Theta(n^2)$?**
A) $5n$
B) $8n^2 + 2n - 3$ ☐
C) $\log n$
D) 2^n
21. **Plane-sweep algorithm improves brute-force by:**
A) Recursion
B) Stack usage ☐
C) Matrix multiplication
D) Sorting edges
22. **Plane-sweep first sorts the points based on:**
A) y-coordinate
B) Weight
C) x-coordinate ☐
D) Value
23. **Which data structure is used in plane-sweep for storing maximal points?**
A) Queue
B) List
C) Tree
D) Stack ☐
24. **Plane-sweep algorithm time complexity:**
A) $\Theta(n^2)$
B) $\Theta(n \log n)$ ☐
C) $\Theta(\log n)$
D) $\Theta(n^3)$
25. **In plane-sweep, why does while-loop execute at most n times?**
A) Due to nested loops

- B) Every element pushed and popped at most once ☐
- C) Sorting limits it
- D) It breaks early

☐ Lecture 6: Asymptotic Comparisons

26. Which is the tightest bound for linear search?

- A) $O(n^2)$
- B) $\Theta(\log n)$
- C) $\Theta(n)$ ☐
- D) $\Theta(1)$

27. Merge sort recurrence relation is:

- A) $T(n) = T(n) + n$
- B) $T(n) = 2T(n/2) + n$ ☐
- C) $T(n) = T(n-1) + \log n$
- D) $T(n) = nT(n-1) + 1$

28. Asymptotic notation helps compare:

- A) Algorithms of same output
- B) Sorting order
- C) Growth rates ☐
- D) Constants

29. When using limits to compare functions, we use:

- A) Derivatives
- B) Recurrences
- C) Limits of $f(n)/g(n)$ ☐
- D) Substitution

30. Which function is not in $\Theta(n^2)$?

- A) $4n^2$
- B) $10n^2$
- C) $n^2 + n$
- D) n^3 ☐

☐ Lecture 7 to 10: Sorting and Strategy

31. Merge Sort time complexity is:

- A) $O(n \log n)$ ☐
- B) $O(n^2)$
- C) $O(n)$
- D) $O(\log n)$

32. Merge Sort is based on which strategy?

- A) Brute force
- B) Divide and Conquer ☐

- C) Dynamic programming
D) Greedy
33. Merge Sort uses which data structure for combining?
A) Heap
B) Queue
C) Temporary array ☐
D) Linked list
34. Merge operation of merge sort takes time:
A) $\Theta(\log n)$
B) $\Theta(n)$ ☐
C) $\Theta(n^2)$
D) $\Theta(1)$
35. Which of the following is not a divide-and-conquer algorithm?
A) Merge Sort
B) Quick Sort
C) Binary Search
D) Counting Sort ☐
-

☐ Lecture 10 Divide and Conquer Continued

36. Which recurrence is solved using recursion tree?
A) $T(n) = nT(n-1)$
B) $T(n) = T(n-1) + 1$
C) $T(n) = 2T(n/2) + n$ ☐
D) $T(n) = T(n/2) + 1$
37. In recursion tree, cost per level is:
A) Decreasing
B) Same ☐
C) Increasing exponentially
D) Always $\log n$
38. Total depth of recursion tree for merge sort is:
A) $\log n$ ☐
B) n
C) $\sqrt{n}$
D) n^2
39. Merge Sort's auxiliary space complexity is:
A) $O(1)$
B) $O(n)$ ☐
C) $O(n^2)$
D) $O(\log n)$
40. Merge sort is preferred over quicksort when:
A) Space is limited
B) Array is small

- C) Stability is required ☐
D) Array is sorted
-

☐ **General Algorithm Knowledge**

41. Which notation gives lower bound only?

- A) O
B) Ω ☐
C) Θ
D) π

42. Which has best-case $\Theta(1)$?

- A) Binary search ☐
B) Merge sort
C) Selection sort
D) Bubble sort

43. Which one is not logarithmic?

- A) $\log n$
B) $\ln n$
C) $n \log n$
D) n^2 ☐

44. Heap sort worst-case time:

- A) $O(n \log n)$ ☐
B) $O(n)$
C) $O(n^2)$
D) $O(\log n)$

45. In divide-and-conquer, base case is when:

- A) Sub-array size is 1 ☐
B) n is even
C) Function calls end
D) Recursion depth is 0

46. Selection problem finds:

- A) Largest element
B) Kth smallest/largest ☐
C) All elements
D) Only even numbers

47. Merge sort is preferred over bubble sort because:

- A) Simpler logic
B) Less space
C) Faster on large datasets ☐
D) Easier to code

48. Which algorithm has the best average-case time for large n ?

- A) Insertion sort
B) Quick sort ☐

- C) Selection sort
- D) Counting sort

49. Which algorithm guarantees $\Theta(n \log n)$ worst-case time?

- A) Merge sort ☒
- B) Quick sort
- C) Heap sort
- D) Bubble sort

50. Which is NOT a divide-and-conquer example?

- A) Merge sort
- B) Binary search
- C) Quick sort
- D) Linear search ☒

learning with Annie